Bilevel Optimal Coordination of Active Distribution Network and Charging Stations Considering EV Drivers' Willingness

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Abstract

With the popularity of electric vehicles (EVs) in urban areas, the effective utilization of their charging flexibility in active distribution networks (ADNs) has drawn wide attentions. This paper proposes a bilevel model for the collaborative operation of ADNs with multiple charging stations (CSs) considering EV drivers' willingness. In the upper level, the distribution system operator (DSO) minimizes the total operational cost of ADNs and sets the optimal energy and reserve prices to trade with CSs under loads and market uncertainties. In the lower level, the aggregated model of EVs including drivers' response willingness is firstly established. Then, CSs minimize their own costs by adjusting bidding quantities and setting optimal incentive prices for EV drivers. Further, the developed bilevel model considering responsivities of EV drivers and various uncertainties is transformed into a tractable single level problem via the deterministic reformulation, Karush-Kuhn–Tucker (KKT) conditions and linearization method. Finally, numerical studies show that the proposed method can facilitate the DSO, CSs, EV drivers to reduce 3.7%, 20%, 9.6% total costs on average and improve the network security of ADNs under various uncertainties. Moreover, the scalability of the proposed approach is verified in the IEEE 123-bus DN with the computation time less than 600s, which satisfies the computational efficiency requirement for the ADN day-ahead optimal scheduling.

Keywords—Active distribution network, electric vehicles, bilevel programming, optimal incentive, response willingness.

NOMENCLATURE

A. Parameters related to EVs and EV aggregator(EVA)

B_i	Battery capacity of EV *i*, in kWh.
$SOC_{\min}$	Minimum State of charge (SOC)
$SOC_{i,d}$	Desired departure SOC of EV *i*.
$SOC_{i,a}$	Arrival SOC of EV *i*.
$p_{i,\max}$	Maximum exchanging power between EV *i* and grid, in kW.
$e_i^{+/-}(t)$	Fastest/slowest charging trajectory of EV *i*, in kWh.
$p_i^{+/-}(t)$	Largest/lowest charging power of EV *i*, in kW.
$E_t^{+/-}$	The EVA upper/lower energy boundary at time *t* in ideal case
$P_t^{+/-}$	The EVA upper/lower power boundary at time *t* in ideal case
η	Charging efficiency of EVs
t / T	Index/set of time intervals in a day
ΔT	Time interval of EV aggregator (1 hour)
Δt	Time interval of an individual EV (5 minute)
TOU	Time of use price for charging stations
$k_{E,t}^{-}$	Gain of EVA lower energy boundary from demand response (DR) of drivers
$k_{P,t}^{-}$	Gain of EVA upper/lower power boundary from DR of drivers

B. Variables related to EVs and EVA

$E_{DR,t}^{+/-}$	The EVA upper/lower energy boundary at time *t* considering DR
$P_{DR,t}^{+/-}$	The EVA upper/lower power boundary at time *t* considering DR
$\pi_t^{P,\,Ru,\,Rd}$	The optimized power/up-reserve/down-reserve prices for CSs

$\lambda_{ev,t}$	The willingness of EVs at time t
$\pi_{inc,t}$	The incentive price at time t
E / $P_{j,t}^{CS}$	The energy/power at j^{th} CS at time t
Ru / $Rd_{j,t}^{CS}$	The up/down-reserve purchased from j^{th} CS at time t

I. Introduction

A. Background and motivation

With the increasing penetration of distributed energy resources (DERs) in the distribution networks (DNs), the traditional DNs are evolving into active DNs (ADNs), which have self-management ability and improve the flexibility and economy of power systems [1]-[3]. However, the intermittency of renewable energy resources (RESs) and uncoordinated energy consumptions might impose negative impacts on the stable operation of ADNs. One promising way to address this problem is to apply coordinated charging of electric vehicles (EVs) [4].

To this end, the increasing penetration of EVs should be coordinated properly and efficiently to provide auxiliary services for ADNs. Besides, to guide EV drivers to participate into demand response (DR) in the ADN, suitable incentives for EV drivers should be designed and the impact of drivers' response willingness on the ADN scheduling should be explored.

B. Literature review

Due to the large quantities of EVs and their uncertainties of spatiotemporal distribution, it is difficult to obtain the optimal charging solution for each individual EV [5]. Therefore, one effective method is to establish the EV aggregator (EVA) model for EV fleets which can dilute the uncertainties of massive EVs and enhance the model's practicability. The power and energy boundary models of diverse EVs are adopted in [6] and the chance constraints program (CCP) is modeled to describe the uncertainties of EVs. Xu *et al.* [7] evaluate the charging flexibility of EV aggregator considering users' willingness. A hybrid incentive program with buy-out (BO) and pay-and-use (PAU) is proposed to increase EV charging flexibility utilization in [8]. Guo *et al.* [9] propose a robust model for EVA considering charging and discharging pressure of EVs. The exact model of EV aggregator is modeled in [10] by the complete binary tree and is applied to optimal dispatching in electricity market. However, these methods are applied to direct EVA participation in the day-ahead market without considering the constraints of ADNs, and lack the active interactions among ADNs, CSs and EV drivers.

To tackle the above drawbacks, the second way for CSs to engage in the electricity market is through the distribution system operator (DSO). CSs first submit their aggregated external characteristics (e.g., the operation boundaries and overall cost functions) to DSO and then DSO computes the optimal scheduling solutions and send the dispatching command to CSs. This is an emerging framework where the local joint energy-reserve market is organized by the DSO [3],[11] and some industrial projects have also been launched in Europe [12].

In recent works, the optimization models between ADN and CSs are usually formulated as bilevel programming problems, where DSO and CSs collaboratively improve the overall economy of the system at different optimization levels with suitable DR programs for EVs [13]. DR refers to balancing the demand on power grids by encouraging customers to shift or adjust electricity demand properly and can be divided into price-based and incentive-based programs [14]. In [3], a collaborative operation model between virtual power plants (VPPs) and ADN with price-based DR is proposed, which sets the optimal prices of active and reactive power for VPPs. Literature [15]-[17] propose bilevel models for combined heat and power systems with price-based DR and the Unscented Transformation method is used to model uncertainties of loads and RESs. In [18], a two-layer optimization model is established to flatten the aggregate load on the power grid and reduce the peak demand of EVs. In [19], the competitive pricing between CSs is considered and the multileader-follower model is formed to reach the Nash equilibrium for all system participants. Park *et al.* [20] propose a bilevel optimization problem to find the optimal DR strategy. Zhang *et al.* [21] design a comprehensive pricing strategy for both energy and ancillary services in ADNs. In [22], a bidirectional electricity trading of DR and transactive energy frameworks between DSO and energy prosumers is proposed. Piltan *et al.*[23] and Pirouzi *et al.*[24] consider resiliency enhancement models of VPPs to achieve the optimal power flow (OPF) for ADNs by stochastic optimization (SO).

Specifically, as for the collaborative optimization of EVs, the bilevel models between EVs and networked microgrids are established in [25], [26] and the incentive-based DR programs are designed to minimize the costs of EVs. In [27] and [28], a bilevel mixed integer nonlinear program (MINP) is formed between EVA and drivers for the optimal reserve management. However, the computational efficiency of these models is low when the number of EVs is large. The optimal energy management for CSs is considered in [29] by the hierarchical game approach. In [30], the linearized OPF of the ADN is obtained through the adjustment of EVA's energy trajectory. In [31], a scenario-based SO model is applied to simulating

different charging patterns of EVs and deciding the optimal bidding of EVA. In [32], the triple-level model for ADNs is proposed to reduce EV charging costs and peak power-grid flows by the robust optimization (RO).

C. Research gaps

Although fruitful achievements for EVs coordination have been in made in aforementioned works, the following issues still remain to be addressed.

Firstly, though the EVA model and EVA participation in the wholesale market have been established in [5]-[10], the network security constraints of ADNs and interactions among ADNs, CSs and EV drivers are ignored.

Secondly, in some of existing works [3] and [15]-[32], the bilevel operation models between ADNs and CSs or VPPs are proposed to reduce ADNs and CSs' costs or improve the resilience of the system. However, the diversified responsivities of EV drivers is neglected. In fact, the optimal scheduling of EVA may degrade drivers' charging satisfactions and drivers might not participate in the optimal scheduling if desired incentives are not provided.

Thirdly, to fill the research gap in considering multiple uncertainties among multiple entities' coordination, the comprehensive optimization model for the interactions among ADNs, CSs and EV drivers with consideration of various uncertainties should be established to improve the profits of system participants as well as the network security.

In a nutshell, the comparisons of EV scheduling with ADNs in existing literature are summarized in Table I as follows.

TABLE I. COMPARISONS OF EV OPTIMAL SCHEDULING IN EXISTING LITERATURE

Category	Ref.	DR program	EV drivers' Responsivity	Uncertainty type	Model formulation
Bidding and scheduling of EVA without considering ADNs	[5]	-	×	1,3	Single-level SO
	[6]	Incentive-based	√	1,3	Single-level CCP
	[7]	Incentive-based	√	3	Single-level SO
	[8]	BO and PAU	×	-	Iterative optimization
	[9]	TOU-based	×	2,3	Single-level RO
Bidding and scheduling of EVA considering ADNs	[3]	Price-based	×	2	Bilevel RO
	[15]-[17]	Price-based	×	1,2	Bilevel SO
	[20]	Incentive-based	√	1,2	Bilevel RO
	[21],[29]	Price-based	×	-	Bilevel iterative optimization
	[23]-[26]	Incentive-based	×	1,2	Bilevel SO
	[27],[28]	Price-based	×	3	Bilevel MINP
	[31]	-	×	2,3	Bilevel SO
	[32]	Incentive-based	×	2,3	Triple-level RO
	Proposed	Hybrid incentive and price-based	√	1,2,3	Bilevel hybrid CCP-SO

* √ : The item is considered; × and -: The item is not considered; 1: electricity market; 2: Load and RESs; 3: EVs.

D. Contributions

Taking the interactions among the ADN, CSs and EV drivers into account, this paper develops a bilevel collaborative operation model between the DSO and EVA operators considering drivers' diversified responsivity to incentive-based DR. The graphical abstract of this article is illustrated in Fig. 1. The main contributions are summarized as follows.

1) This paper considers diversified responsivity of EV drivers to participate in the bilevel optimization based on consumer psychology model and designs an incentive-based DR for EVs. Then, the response capabilities (RCs) of EVs with diversified responsivity are further modeled by the aggregated model and embedded into the bilevel optimization.

2) The bilevel optimization model for interactions among the ADN, CSs and EV drivers considering various uncertainties is established to improve the profits of system participants and reduce constraint violations. Besides, the comprehensive uncertainties in the system are modeled by a hybrid CCP-SO to improve the security operation of the ADN. Numerical studies show that the proposed method can facilitate the DSO, CSs, EV drivers to reduce 3.7%, 20%, 9.6% total costs on average.

3) The tractable reformulation of the proposed bilevel model considering various uncertainties of ADNs and different responsivity types of EV drivers is developed via the deterministic reformulation, Karush-Kuhn–Tucker (KKT) conditions and linearization method, which is computationally efficient for the ADN day-ahead optimization.

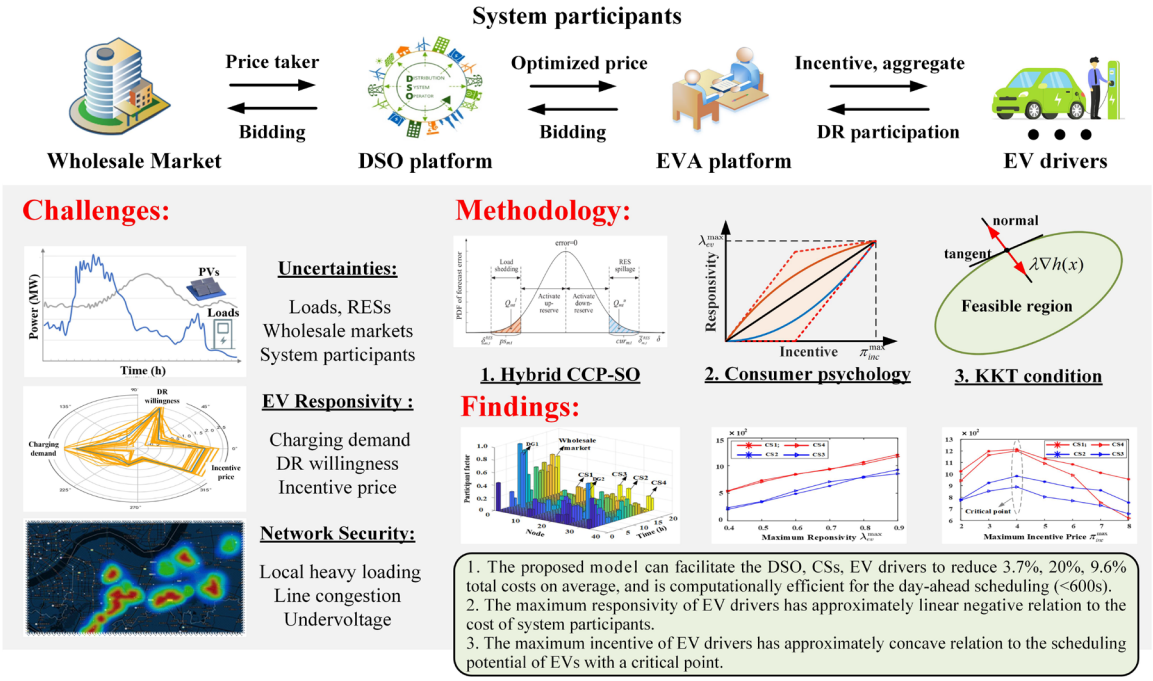


Fig. 1. The graphical abstract of proposed bilevel optimal coordination of ADNs and CSs considering drivers' willingness.

II. Problem Statement and Bilevel Framework

In the traditional distribution system, there is little elastic load and DSO only broadcasts the energy prices to the load users. However, with the increasing integration of DERs in the ADN, DERs can provide energy and ancillary services back to ADN and improve the economy of the whole system. Therefore, massive DERs are encouraged to interact with DSO to sell their scheduling flexibility and ancillary services.

Specifically, this paper focus on the scheduling of public CSs in ADNs considering EV drivers' willingness. The enhanced controllability of public CSs can help ADN actively regulate the EV charging to improve the stable and economic operation of ADN. Besides, to fully realize scheduling potential of EVs, the CSs need to offer suitable incentives to EV drivers for DR and aggregate the information of EVs. In a word, DSO acts as a mediator and is responsible for trading with the wholesale market as well as interacting with CSs.

A. Model assumptions

Some preliminary assumptions of the proposed model are given as follows:

1) The ADN and public CSs are assumed to have different interests. Each market participant has its own objective function and seeks to minimize its cost. Moreover, the ADN and CSs operate collaboratively to reach the optima simultaneously. Thus, the proposed model assumes that the information of CSs submitted to DSO is accurate and adequate, as in [15]-[32].

2) The DSO is assumed to participate in the day-ahead wholesale electricity markets as a price taker. Besides, the wholesale energy and reserve prices are predicted using historical data, since the main focus of this paper is the collaborative optimization and pricing design for DSO and CSs, which is similar to [15]-[32].

3) The prediction errors of RESs are assumed to obey the Gaussian distribution with the known variance, which is consistent with [3]-[5].

B. Bilevel optimization framework

In this paper, the bidding and pricing strategy between ADN and CSs is modeled as a bilevel optimization. Besides, the incentives for EV drivers' participation in scheduling is designed to exploit the scheduling potential of EVs. The illustration of the bilevel framework is shown in Fig. 2. Different levels have their own objective functions and variables, and they interact with each other by transferring the coupled variables.

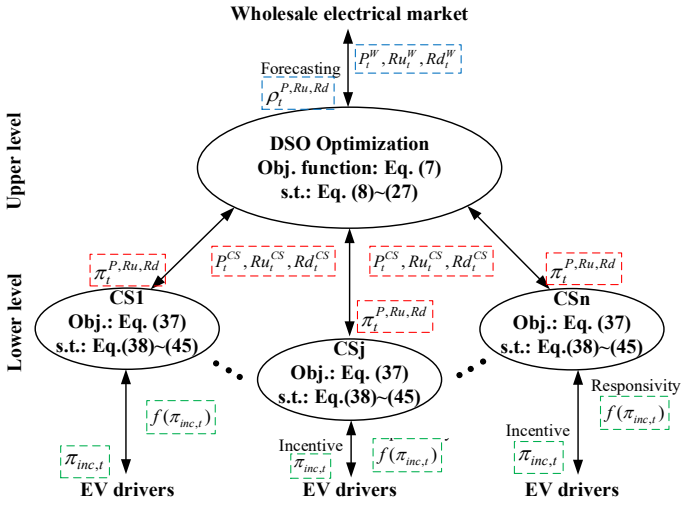


Fig. 2. The framework of the bilevel optimization between the DSO and CSs.

At the upper level, DSO is responsible to optimize the total cost of ADN with consideration of reserve capacity sufficiency and network constraints. Meanwhile, the decision variables of DSO include scheduling plans for its internal sources (e.g., distributed generators (DGs) and RESs owned by DSO, bidding plans in the wholesale energy and reserve markets, and the energy and reserve prices for CSs at different locations.

At the lower level, different CSs seek to minimize their total cost after receiving the prices issued by DSO. Moreover, CSs need to decide the incentive prices for EV drivers who respond to the DR program. Therefore, the decision variables of CSs include the energy and reserve bids/offers, and incentive prices for EV drivers.

III. Mathematic Model of Upper-level Optimization

A. The uncertainty models in ADNs and wholesale market

Unlike controllable DGs, RESs and loads inherit stochastic nature. In this regard, the flexible reserve should be determined properly to deal with power imbalance incurred by uncertainties from RESs and loads. Define the forecast net load at node m as $P_m^{L,net} = P_m^L - P_m^{RES}$. Then, the practical net load can be modeled as

$$\tilde{P}_{m,t}^{L,net} = P_{m,t}^{L,net} + \xi_{m,t}^{L,net} \tag{1}$$

where $\xi_{m,t}^{L,net} = \xi_{m,t}^{L} - \xi_{m,t}^{RES}$ is the corresponding forecast error from RESs and loads. In accordance with [3]-[6], $\xi_{m,t}^{L,net}$ is assumed to be normally distributed. Further, the total forecast error in the ADN is

$$\zeta_t = \sum\nolimits_m \xi_{m,t}^{L,net} = \mathbf{1}^{\mathrm{T}} \boldsymbol{\xi}_t^{L,net} \tag{2}$$

where $\mathbf{1}$ is the column vector of all ones. $\boldsymbol{\xi}_t^{L,net}$ is the random vector for $\xi_{m,t}^{L,net}$, with the expected value and covariance matrix given by

$$\mathbb{E}\left[\boldsymbol{\xi}_t^{L,net}\right] = \mathbf{0}, \sigma^2\left[\boldsymbol{\xi}_t^{L,net}\right] = \Sigma_t \tag{3}$$

The DGs and controllable loads (i.e., EVs) can adjust their power trajectories and provide reserves to offset these uncertainties and maintain the power balance of the ADNs. By the affine control policy [33], the power outputs of the wholesale market, DGs, and CSs considering uncertainties can be modeled as

$$\tilde{P}_t^W = P_t^W + \alpha_{0,t}\zeta_t \tag{4}$$

$$\tilde{P}_{i,t}^G = P_{i,t}^G + \alpha_{i,t}\zeta_t; \forall i \in N_G \tag{5}$$

$$\tilde{P}_{j,t}^{CS} = P_{j,t}^{CS} - \beta_{j,t}\zeta_t; \forall j \in N_{CS} \tag{6}$$

where $\left\{P_t^W, P_{i,t}^G, P_{j,t}^{CS}\right\}$ are expected power outputs from the wholesale market, DGs and CSs at time t, and $\left\{\tilde{P}_t^W, \tilde{P}_{i,t}^G, \tilde{P}_{j,t}^{CS}\right\}$ are disturbed power outputs considering uncertainties. N_G, N_{CS} are the sets of DGs and CSs. $\alpha_{0,t}, \alpha_{i,t}, \beta_{j,t}$ are the corresponding participant factors, satisfying $\alpha_{0,t} + \sum_i \alpha_{i,t} + \sum_j \beta_{j,t} = 1; \forall t \in T$ to enforce the power balance of ADNs under uncertainties of loads and RESs.

Apart from power imbalance uncertainties, the uncertainties also exist in the day-ahead wholesale market (i.e., the energy and reserve prices) [15]-[17]. The DSO operator should make the optimal bidding strategy to minimize its expected cost from the wholesale market among typical scenarios which is illustrated in Eq. (7).

B. Objective function of upper-level optimization

The DSO operator aims to minimize the total operational cost consisting of: i). Generation and reserve cost of DGs, ii). Cost of reserve service subtracting revenue of EV charging from CSs, and iii). The expected energy and reserve cost from wholesale market. Then the objective function of DSO can be formed and expressed as

$$\min_{x^{DSO}} f_0^{DSO}\left(x^{DSO}\right)=\sum_{t=1}^{T}\sum_{i\in N_G} C_i^G\left(P_{i,t}^G,Ru_{i,t}^G,Rd_{i,t}^G\right)+\sum_{t=1}^{T}\sum_{j\in N_{CS}} C_j^{CS}\left(P_{j,t}^{CS},Ru_{j,t}^{CS},Rd_{j,t}^{CS}\right)+\sum_s \gamma_s \sum_{t=1}^{T} C^W\left(P_t^W,Ru_t^W,Rd_t^W\right) \tag{7}$$

$$C_i^G\left(P_{i,t}^G,Ru_{i,t}^G,Rd_{i,t}^G\right)=cg_{1,i}\left(P_{i,t}^G\right)^2+cg_{2i}P_{i,t}^G+cr_i^+Ru_{i,t}^G+cr_i^-Rd_{i,t}^G \tag{8}$$

$$C^W\left(P_t^W,Ru_t^W,Rd_t^W\right)=\rho_{s,t}^P P_t^W+\rho_{s,t}^{Ru}Ru_t^W+\rho_{s,t}^{Rd}Rd_t^W \tag{9}$$

$$C_j^{CS}\left(P_{j,t}^{CS},Ru_{j,t}^{CS},Rd_{j,t}^{CS}\right)=\pi_{j,t}^{Ru}Ru_{j,t}^{CS}+\pi_{j,t}^{Rd}Rd_{j,t}^{CS}-\pi_{j,t}^{P}P_{j,t}^{CS}. \tag{10}$$

where $x_t^{DSO}=\left\{P_{i,t}^G,Ru_{i,t}^G,Rd_{i,t}^G,\alpha_{i,t},P_t^W,Ru_t^W,Rd_t^W,\alpha_{0,t},\pi_{j,t}^P,\pi_{j,t}^{Ru},\pi_{j,t}^{Rd},\beta_{j,t}\right\}$ are upper-level decision variables for $\forall t\in T$, $\forall i\in N_{DG},\forall j\in N_{CS}$. $\left\{Ru,Rd\right\}$ denote the up/down-reserves of dispatchable sources. $\left\{cg_{1,i},cg_{2i}\right\}$ denote power cost coefficients of DG$_i$. $\left\{cr_i^+,cr_i^-\right\}$ are the up/down-reserve cost coefficients of DG$_i$. $\left\{\rho_{s,t}^P,\rho_{s,t}^{Ru},\rho_{s,t}^{Rd}\right\}$ are the power/up-reserve/down-reserve prices in the wholesale market at scenario s. γ_s is the probability of the scenario s to happen.

C. Constraints of upper-level optimization

Several constraints of DSO are considered as follows. Firstly, the linearized power flow model for distribution network is implemented as follows [4]:

$$P_{mn,t}=g_{mn}\left(V_{m,t}-V_{n,t}\right)-b_{mn}\left(\theta_{m,t}-\theta_{n,t}\right) \tag{11}$$

$$Q_{mn,t}=-b_{mn}\left(V_{m,t}-V_{n,t}\right)-g_{mn}\left(\theta_{m,t}-\theta_{n,t}\right) \tag{12}$$

where $g_{mn}\geq 0$ and $b_{mn}\leq 0\ \left(m\neq n\right)$ are the conductance and susceptance between node m and n, respectively. Then, the power balance at each node should be satisfied:

$$p_{m,t}=\sum_{i=1}^{N_B}P_{mn,t}=P_{m,t}^G+P_t^W-P_{m,t}^{L,net}-P_{m,t}^{CS} \tag{13}$$

$$q_{m,t}=\sum_{i=1}^{N_B}Q_{mn,t}=Q_{m,t}^G+Q_t^W-Q_{m,t}^{L,net} \tag{14}$$

where N_B is the number of buses in the ADN. Besides, the power flow model in Eq. (11)-(14) can be further written in the compact form as

$$\begin{aligned}\mathbf{p}_t&=-\mathbf{B}\boldsymbol{\theta}_t+\mathbf{G}\mathbf{V}_t\\ \mathbf{q}_t&=-\mathbf{B}\mathbf{V}_t-\mathbf{G}\boldsymbol{\theta}_t\end{aligned} \tag{15}$$

where $\mathbf{G},\mathbf{B}$ are the reduced invertible admittance matrix without the slack bus at the substation. The limits of power generation of DGs and power transmission between the ADN and substation are given by:

$$P_{i,t}^G-Rd_{i,t}^G\geq P_{i,\min}^G,P_{i,t}^G+Ru_{i,t}^G\leq P_{i,\max}^G,Q_{i,\min}^G\leq Q_{i,t}^G\leq Q_{i,\max}^G;\forall i\in N_G \tag{16}$$

$$P_t^W-Rd_t^W\geq P_{\min}^W,P_t^W+Ru_t^W\leq P_{\max}^W,Q_{\min}^W\leq Q_t^W\leq Q_{\max}^W \tag{17}$$

where $P_{i,\max}^G/Q_{i,\max}^G,P_{i,\min}^G/Q_{i,\min}^G$ are the upper and lower limits of power generation of DGs. $P_{\max}^W/Q_{\max}^W,P_{\min}^W/Q_{\min}^W$ are the bidding limits from the wholesale market.

To offset the uncertainties from RESs and loads, the following chance constraints based on Eq. (1)-(6) are modeled.

$$\Pr_{\xi_t^{L,net}\sim N(0,\Sigma_t)}\left\{\alpha_{i,t}\zeta_t\leq Ru_{i,t}^{G/W}\right\}\geq 1-\varepsilon;\ \Pr_{\xi_t^{L,net}\sim N(0,\Sigma_t)}\left\{-\beta_{i,t}\zeta_t\leq Ru_{i,t}^{CS}\right\}\geq 1-\varepsilon \tag{18}$$

$$\Pr_{\xi_t^{L,net}\sim N(0,\Sigma_t)}\left\{-\alpha_{i,t}\zeta_t\leq Rd_{i,t}^{G/W}\right\}\geq 1-\varepsilon;\ \Pr_{\xi_t^{L,net}\sim N(0,\Sigma_t)}\left\{\beta_{i,t}\zeta_t\leq Rd_{i,t}^{CS}\right\}\geq 1-\varepsilon \tag{19}$$

where ε is the violation probability chosen by the DSO. Meanwhile, the network security of nodal voltage and power flow in the ADN considering uncertainties from RESs and loads also should be guaranteed in a desired confidence level. Therefore, the following chance constraints are given by

$$\Pr_{\xi_t^{L,net}\sim N(0,\Sigma_t)}\left\{-\tilde{V}_{m,t}\le -V_{\min}\right\}\ge 1-\varepsilon;\ \Pr_{\xi_t^{L,net}\sim N(0,\Sigma_t)}\left\{\tilde{V}_{m,t}\le V_{\max}\right\}\ge 1-\varepsilon \tag{20}$$

$$\Pr_{\xi_t^{L,net}\sim N(0,\Sigma_t)}\left\{-\tilde{P}_{mn,t}\le -P_{mn}^{\min}\right\}\ge 1-\varepsilon;\ \Pr_{\xi_t^{L,net}\sim N(0,\Sigma_t)}\left\{\tilde{P}_{mn,t}\le P_{mn}^{\max}\right\}\ge 1-\varepsilon \tag{21}$$

where $\tilde{V}_{m,t},\tilde{P}_{mn,t}$ are the nodal voltage and power flow under uncertainties from RESs and loads, respectively. $V_{\max},V_{\min}$ are the upper and lower limits of nodal voltage in the ADN. $P_{mn}^{\max},P_{mn}^{\min}$ are the capacity limits of power transmission at line *mn*. To explicitly incorporate ξ_t^{UL} into $\tilde{V}_{m,t},\tilde{P}_{mn,t}$, according to Eq. (15), $\tilde{V}_{m,t}$ can be firstly derived as

$$\tilde{\mathbf{V}}_t=\underbrace{\left(\mathbf{G}+\mathbf{B}\mathbf{G}^{-1}\mathbf{B}\right)^{-1}}_{\mathbf{R}}\tilde{\mathbf{p}}_t+\underbrace{\left(-\mathbf{B}-\mathbf{G}\mathbf{B}^{-1}\mathbf{G}\right)^{-1}}_{\mathbf{X}}\mathbf{q}_t=\mathbf{V}_t+\mathbf{R}\left(\zeta_t^{UL}\boldsymbol{\alpha}_t+\zeta_t^{UL}\boldsymbol{\beta}_t-\boldsymbol{\xi}_t^{L,net}\right) \tag{22}$$

In addition, considering the ADNs usually have the radial topology [34], the relationship between power flow and nodal power can be simplified as

$$\mathbf{P}_t=\mathbf{D}\mathbf{p}_t,\mathbf{Q}_t=\mathbf{D}\,\mathbf{q}_t \tag{23}$$

where $\mathbf{D}$ is the downstream node-to-node (as well as line-to-line) incidence matrix of the ADNs [34]. Thereby, $\tilde{P}_{mn,t}$ can be derived as

$$\tilde{\mathbf{P}}_t=\mathbf{P}_t+\mathbf{D}\left(\zeta_t^{UL}\boldsymbol{\alpha}_t+\zeta_t^{UL}\boldsymbol{\beta}_t-\boldsymbol{\xi}_t^{L,net}\right) \tag{24}$$

Substituting Eq. (22) and (24) into (20) and (21), the chance constraints of network security constraints can be reformulated as

$$\Pr_{\xi_t^{L,net}\sim N(0,\Sigma_t)}\left\{-\mathbf{R}\left(\zeta_t^{UL}\boldsymbol{\alpha}_t+\zeta_t^{UL}\boldsymbol{\beta}_t-\boldsymbol{\xi}_t^{L,net}\right)\le \mathbf{V}_t-\mathbf{V}_{\min}\right\}\ge 1-\varepsilon;\ \Pr_{\xi_t^{L,net}\sim N(0,\Sigma_t)}\left\{\mathbf{R}\left(\zeta_t^{UL}\boldsymbol{\alpha}_t+\zeta_t^{UL}\boldsymbol{\beta}_t-\boldsymbol{\xi}_t^{L,net}\right)\le \mathbf{V}_{\max}-\mathbf{V}_t\right\}\ge 1-\varepsilon \tag{25}$$

$$\Pr_{\xi_t^{L,net}\sim N(0,\Sigma_t)}\left\{-\mathbf{D}\left(\zeta_t^{UL}\boldsymbol{\alpha}_t+\zeta_t^{UL}\boldsymbol{\beta}_t-\boldsymbol{\xi}_t^{L,net}\right)\le \mathbf{P}_t-\mathbf{P}_{\min}\right\}\ge 1-\varepsilon;\ \Pr_{\xi_t^{L,net}\sim N(0,\Sigma_t)}\left\{\mathbf{D}\left(\zeta_t^{UL}\boldsymbol{\alpha}_t+\zeta_t^{UL}\boldsymbol{\beta}_t-\boldsymbol{\xi}_t^{L,net}\right)\le \mathbf{P}_{\max}-\mathbf{P}_t\right\}\ge 1-\varepsilon\,. \tag{26}$$

Finally, the power and reserve prices for CSs offered by the DSO should maintain within a reasonable range to motivate CSs to respond to the scheduling spontaneously. Thereby, the following constraints are given by

$$\pi_{\min}^{\{\cdot\}}\le\pi_t^{\{\cdot\}}\le\pi_{\max}^{\{\cdot\}};\{\cdot\}=\{P,Ru,Rd\}\,. \tag{27}$$

IV. Mathematic Model of Lower-level Optimization Considering Drivers' Willingness

A. The aggregated model of CS operation range

It is difficult to predict the accurate temporal distribution of individual EV due to the stochastic charging behavior of EV drivers. Therefore, one effective method is to establish the EVA model for EV fleets which can enhance the model's practicability.

The charging data recorded at CSs include the arrival time (t_a), departure time (t_d), state of charge (SOC) at the arrival time ($SOC_{i,a}$), and desired SOC ($SOC_{i,d}$). Specifically, the real charging data at CSs are used to establish the model of CS operation range as follows:

Firstly, the operation region of a single EV data can be modeled by its energy and power boundaries, i.e., $\left\{e_i^{+/-}(t),p_i^{+/-}(t)\right\}$ with its arrival and departure time, i.e., $\{t_a,t_d\}$. Considering the battery lifetime and driver satisfaction [35], only unidirectional V2G operation is permitted and the following two parameters of EV *i* are defined first.

$$\begin{cases}e_{i,d}=B_i\left(SOC_{i,d}-SOC_{i,a}\right)\\ e_{i,\min}=B_i\left(SOC_{\min}-SOC_{i,a}\right)\end{cases}. \tag{28}$$

Secondly, the charging EVs are divided into controllable EVs and uncontrollable EVs. For uncontrollable EVs which can't be charged to $SOC_{i,d}$ during parking, the area of feasible region is zero (i.e., $e_i^+=e_i^-$, $p_i^+=p_i^-$). For controllable EVs, the operation boundaries are described as follows.

$$e_i^+(t)=\begin{cases}0,t<t_a\\ \min\left(p_{i,\max}\eta\left(t-t_a\right),e_{i,d}\right),t_a\le t\le t_d\\ e_i^+\left(t_d\right),t>t_d\end{cases} \tag{29}$$

$$e_i^-(t)=\begin{cases}0, t<t_a\\ \max\left(e_{i,\min}, e_{i,d}-p_{i,\max}\eta\left(t_d-t\right)\right), t_a\le t\le t_d\\ e_i^-\left(t_d\right), t>t_d\end{cases}. \tag{30}$$

Meanwhile, the power boundaries $p_i^{+/-}(t)$ are given by

$$p_i^+(t)=\begin{cases}0, t<t_a \text{ or } t>t_d\\ \min\left(\dfrac{e_i^+(t+\Delta t)-e_i^-(t)}{\eta\ \Delta t}, p_{i,\max}\right), t_a\le t\le t_d\end{cases} \tag{31}$$

$$p_i^-(t)=\begin{cases}0, t<t_a \text{ or } t>t_d\\ \max\left(\dfrac{e_i^-(t+\Delta t)-e_i^+(t)}{\eta\ \Delta t}, 0\right), t_a\le t\le t_d\end{cases}. \tag{32}$$

After establishing the models of individual EVs, the feasible energy and power ranges of EVA are calculated by

$$\begin{cases}E_t^-=\sum_{i\in N}e_i^-(t)\le E_t^{CS}\le\sum_{i\in N}e_i^+(t)=E_t^+\\ P_t^-=\sum_{i\in N}p_i^-(t)\le P_t^{CS}\le\sum_{i\in N}p_i^+(t)=P_t^+\\ \eta P_t^{CS}\Delta T=E_{t+1}^{CS}-E_t^{CS}\end{cases}. \tag{33}$$

B. Drivers' response willingness

EVs have a considerable flexibility in optimal scheduling as shown in (33). However, the uncoordinated EV charging (i.e. E_t^{CS} follows the trajectory of E_t^+ to charge) at public CSs can have a negative effect on the stable operation of ADNs. Hence, it is necessary to set incentives for EV drivers and encourage them to sell their charging flexibility.

Firstly, the response willingness of total EVs at time *t* to the incentive price of DR participation can be modeled as a linear function of incentive price $\pi_{inc,t}$:

$$\lambda_{ev,t}=\begin{cases}\lambda_{ev}^{\max}\times\pi_{inc,t}/\pi_{inc}^{\max}, \pi_{inc,t}\le\pi_{inc}^{\max}\\ \lambda_{ev}^{\max}, \pi_{inc,t}>\pi_{inc}^{\max}\end{cases} \tag{34}$$

where $\lambda_{ev}^{\max}<1$ is the maximum responsivity of drivers. $\pi_{inc}^{\max}$ is the maximum incentive corresponding to $\lambda_{ev}^{\max}$. In practice, $\pi_{inc}^{\max}$ is the parameter determined by EV drivers' willingness of DR participation (e.g. $\pi_{inc}^{\max}$ will be lower if EV drivers have higher willingness to respond to DR.) The influence of $\pi_{inc}^{\max}$ will be carefully explored in Section V.A.3.

Meanwhile, the behavior of drivers is also uncertain because of different sensitivity to incentive price and traveling needs. In [7] and [36], the deviation range of DR is modeled by triangle membership function based on consumer psychology model, as shown in Fig. 3, where π_{inc}^{IP} is the inflection point. Considering the dilution effect of EVA on EVs' uncertainties and making the trade-off between reliability and benefits for CSs, the fitting function for drivers' willingness ($\lambda_{ev,t}$) is defined as $f\left(\pi_{inc,t}\right)$. The range of $f\left(\pi_{inc,t}\right)$ is smaller than the upper boundary and larger than the lower boundary of willingness. Both convex and concave functions can be used for curving fitting of drivers' willingness which depends on the practical statistics and typical scenarios. Some illustrative fitting curves of $\lambda_{ev,t}$ are presented in Fig. 3.

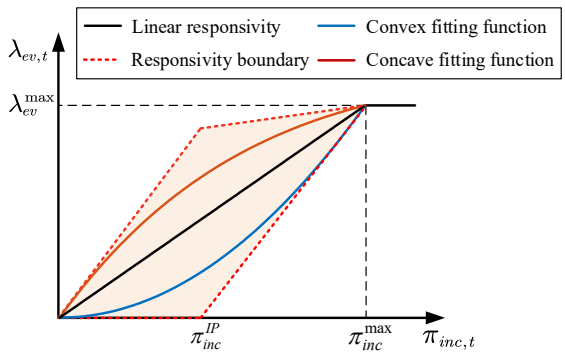


Fig. 3. The curves of price-based willingness.

Furthermore, the response willingness of EV drivers will influence the feasible area of energy and power region, and larger responsivity of drivers can improve the available EV capacity. On the other hand, RCs of different EVs vary considerably due to different charging (or parking) duration. To coordinate EVs with different RCs, the **hourly RC incentive program** is proposed in this paper. More specifically, EV drivers can respond to the incentive $\pi_{inc,t}$ during their charging and $\pi_{inc,t}$ denotes the incentive for hourly DR participation of each EV driver. Define $T_{DR,t} \in \mathbb{Z}$ as the maximum effective response time among EVs at time period *t*. The total incentives for EVs' DR at time period *t* is equal $\lambda_{ev,t} T_{DR,t} \pi_{inc,t}$. Meanwhile, the energy boundary and power boundary of EVA considering the willingness of drivers can be expressed as

$$\begin{cases} E_{DR,t}^{+} = E_t^{+} \\ E_{DR,t}^{-} = E_t^{+} - k_{E,t}^{-} f\left(\pi_{inc,t}\right) T_{DR,t} \\ P_{DR,t}^{+} = P_{un,t} + k_{P,t}^{+} f\left(\pi_{inc,t}\right) T_{DR,t} \\ P_{DR,t}^{-} = P_{un,t} - k_{P,t}^{-} f\left(\pi_{inc,t}\right) T_{DR,t} \end{cases} \tag{35}$$

where $P_{un,t} = \left(E_{t+1}^{+} - E_t^{+}\right) / \left(\eta \Delta T\right)$ is the uncoordinated charging power of EVA. Parameters $k_{E,t}^{-}, k_{P,t}^{+/-}$ can be calculated by

$$\begin{cases} k_{E,t}^{-} = \left(E_t^{+} - E_t^{-}\right) / T_{DR,t} \\ k_{P,t}^{+} = \left(P_t^{+} - P_{un,t}\right) / T_{DR,t} \\ k_{P,t}^{-} = \left(P_{un,t} - P_t^{-}\right) / T_{DR,t} \end{cases}. \tag{36}$$

C. Lower-level optimization: optimal scheduling of CSs considering drivers' willingness

The CS operators aim to minimize its own cost and determine the biding of energy and reserve according to the prices offered by the DSO. Besides, CSs need to set the optimal incentive price for drivers who respond to the proposed DR program. Thereby, the objective function of jth CS can be established as:

$$\min_{x_j^{CS}} f_0^{CS}\left(x_j^{CS}\right) = \sum_{t=1}^{T} \left(\pi_{j,t}^{P} P_{j,t}^{CS} + T_{DRj,t} f\left(\pi_{incj,t}\right) \pi_{incj,t} - \pi_{j,t}^{Ru} Ru_{j,t}^{CS} - \pi_{j,t}^{Rd} Rd_{j,t}^{CS} \right) \tag{37}$$

where $x_j^{CS} = \left\{ P_{j,t}^{CS}, E_{j,t}^{CS}, Ru_{j,t}^{CS}, Rd_{j,t}^{CS}, \pi_{incj,t} \right\}$ are the lower-level decision variables for $\forall t \in T, \forall j \in N_{CS}$.

Several constraints of CSs are considered as follows. Firstly, the equality constraint between energy and power of EVA is

$$E_{j,t}^{CS} = E_{j,t-1}^{CS} + \eta P_{j,t-1}^{CS} \Delta T : \nu_{1j,t}, \forall t, j . \tag{38}$$

Meanwhile, the total charging demand at CSs in one day should be satisfied and is expressed as:

$$E_j^{CS}\left(t = T\right) = E_j^{+}\left(t = T\right) : \nu_{2j}, \forall j . \tag{39}$$

Besides, the power, reserve and energy limit considering drivers' willingness are given by

$$P_{DRj,t}^{+} - P_{j,t}^{CS} - Rd_{j,t}^{CS} \geq 0 : \overline{\lambda}_{2j,t}, \forall t, j \tag{40}$$

$$P_{j,t}^{CS} - Ru_{j,t}^{CS} - P_{DRj,t}^{-} \geq 0 : \underline{\lambda}_{2j,t}, \forall t, j \tag{41}$$

$$Ru_{j,t}^{CS} \geq 0 : \underline{\lambda}_{3j,t}, \forall t, j \tag{42}$$

$$Rd_{j,t}^{CS} \geq 0 : \underline{\lambda}_{4j,t}, \forall t, j \tag{43}$$

$$E_{DRj,t}^{+} - E_{j,t}^{CS} \geq 0,\ E_{j,t}^{CS} - E_{DRj,t}^{-} \geq 0 : \overline{\lambda}_{5j,t}, \underline{\lambda}_{5j,t}\ \forall t, j \tag{44}$$

$$\pi_{inc}^{\max} - \pi_{incj,t} \geq 0, \pi_{incj,t} \geq 0 : \overline{\lambda}_{6j,t}, \underline{\lambda}_{6j,t}, \forall t, j . \tag{45}$$

V. Model Reformulation and Solution Methodology

A. Deterministic reformulation of chance constraints

The uncertainties from RESs and loads are modeled by chance constraints in Eqs. (18), (19), (25), (26) and can be rewritten in a general form as follows.

$$\Pr_{\xi_t^{L,net} \sim N(0,\Sigma_t)} \left\{ a_i\left(x_t^{DSO}\right)^{\mathrm{T}} \xi_t^{L,net} \leq b_i\left(x_t^{DSO}\right) \right\} \geq 1 - \varepsilon \tag{46}$$

where index i denotes the i-th chance constraint. Considering the forecast errors $\xi_t^{L,net}$ follow the normal distribution as depicted in (3), Eq. (46) can be reformulated as the following analytic expression:

$$b_i\left(x_t^{DSO}\right)-\Phi^{-1}\left(1-\varepsilon\right)\left\|a_i\left(x_t^{DSO}\right)^{\mathrm{T}}\Sigma_t a_i\left(x_t^{DSO}\right)\right\|_2\geq 0 \tag{47}$$

where $\Phi^{-1}(\cdot)$ is the inverse Gaussian cumulative distribution. Based on Eq. (47), exact reformulations of Eqs. (18), (19), (25), (26) can be derived as follows:

$$Ru_{i,t}^{G/W}-\Phi^{-1}\left(1-\varepsilon\right)\cdot\mathrm{tr}\left(\Sigma_t\right)\cdot\alpha_{i,t}\geq 0;\ Ru_{i,t}^{CS}-\Phi^{-1}\left(1-\varepsilon\right)\cdot\mathrm{tr}\left(\Sigma_t\right)\cdot\beta_{i,t}\geq 0 \tag{48}$$

$$Rd_{i,t}^{G/W}-\Phi^{-1}\left(1-\varepsilon\right)\cdot\mathrm{tr}\left(\Sigma_t\right)\cdot\alpha_{i,t}\geq 0;\ Rd_{i,t}^{CS}-\Phi^{-1}\left(1-\varepsilon\right)\cdot\mathrm{tr}\left(\Sigma_t\right)\cdot\beta_{i,t}\geq 0 \tag{49}$$

$$\mathbf{V}_t-\mathbf{V}_{\min}-\Phi^{-1}\left(1-\varepsilon\right)\mathbf{R}\left(\mathrm{tr}\left(\Sigma_t\right)\boldsymbol{\alpha}_t+\mathrm{tr}\left(\Sigma_t\right)\boldsymbol{\beta}_t+\Sigma_t\mathbf{1}\right)\geq 0 \tag{50}$$

$$\mathbf{V}_{\max}-\mathbf{V}_t-\Phi^{-1}\left(1-\varepsilon\right)\mathbf{R}\left(\mathrm{tr}\left(\Sigma_t\right)\boldsymbol{\alpha}_t+\mathrm{tr}\left(\Sigma_t\right)\boldsymbol{\beta}_t+\Sigma_t\mathbf{1}\right)\geq 0 \tag{51}$$

$$\mathbf{P}_t-\mathbf{P}_{\min}-\Phi^{-1}\left(1-\varepsilon\right)\mathbf{D}\left(\mathrm{tr}\left(\Sigma_t\right)\boldsymbol{\alpha}_t+\mathrm{tr}\left(\Sigma_t\right)\boldsymbol{\beta}_t+\Sigma_t\mathbf{1}\right)\geq 0 \tag{52}$$

$$\mathbf{P}_{\max}-\mathbf{P}_t-\Phi^{-1}\left(1-\varepsilon\right)\mathbf{D}\left(\mathrm{tr}\left(\Sigma_t\right)\boldsymbol{\alpha}_t+\mathrm{tr}\left(\Sigma_t\right)\boldsymbol{\beta}_t+\Sigma_t\mathbf{1}\right)\geq 0 \tag{53}$$

where $\mathrm{tr}\left(\Sigma_t\right)$ is the trace of Σ_t.

B. Equivalent single-level model

The proposed bilevel program can be transformed into a single-level problem by incorporating the lower-level optimization problem into the upper-level problem with Karush-Kuhn–Tucker (KKT) optimality condition. Firstly, rewrite constraints (38)-(45) as $\left\{h_k^{CS}\left(x_j^{CS}\right)=0,k\in N_{eq}\right\}$, and $\left\{f_i^{CS}\left(x_j^{CS}\right)\leq 0,i\in N_{ineq}\right\}$. N_{eq} is the set of equality (38)-(39) and N_{ineq} is the set of inequality (40)-(45). Then the Lagrange function of the lower-level problem can be constructed by combining the objective function and constraints as follows:

$$L\left(x_j^{CS},\lambda,\nu\right)=f_0^{CS}\left(x_j^{CS}\right)+\sum_{k\in N_{eq}}\nu_k h_k^{CS}\left(x_j^{CS}\right)+\sum_{i\in N_{ineq}}\lambda_i f_i^{CS}\left(x_j^{CS}\right). \tag{54}$$

Assume h_k^{CS}, f_i^{CS} are differentiable, the KKT conditions of jth CS can be derived as follows:

$$\boldsymbol{\pi}_j^{\mathbf{P}}+\overline{\boldsymbol{\lambda}}_{2j}-\underline{\boldsymbol{\lambda}}_{2j}+\left(\mathbf{A}^{-1}\right)^{\mathbf{T}}\left(\overline{\boldsymbol{\lambda}}_{5j}-\underline{\boldsymbol{\lambda}}_{5j}\right)+\nu_{2j}\mathbf{1}=0 \tag{55}$$

$$-\boldsymbol{\pi}_j^{Ru}+\underline{\boldsymbol{\lambda}}_{2j}-\underline{\boldsymbol{\lambda}}_{3j}=0 \tag{56}$$

$$-\boldsymbol{\pi}_j^{Rd}+\overline{\boldsymbol{\lambda}}_{2j}-\underline{\boldsymbol{\lambda}}_{4j}=0 \tag{57}$$

$$\begin{aligned}&\overline{\boldsymbol{\lambda}}_{6j}-\underline{\boldsymbol{\lambda}}_{6j}+\mathbf{T}_{\mathbf{DR}j}\cdot\left(\mathbf{f}\left(\boldsymbol{\pi}_{\mathbf{inc}j}\right)+\frac{\mathbf{df}\left(\boldsymbol{\pi}_{\mathbf{inc}j}\right)}{\mathbf{d}\boldsymbol{\pi}_{\mathbf{inc}j}}\boldsymbol{\pi}_{\mathbf{inc}j}\right)\\&=\mathbf{k}_{\mathbf{P}j}^{+}\cdot\mathbf{T}_{\mathbf{DRj}}\cdot\frac{\mathbf{df}\left(\boldsymbol{\pi}_{\mathbf{inc}j}\right)}{\mathbf{d}\boldsymbol{\pi}_{\mathbf{inc}j}}\overline{\boldsymbol{\lambda}}_{2j}+\mathbf{k}_{\mathbf{P}j}^{-}\cdot\mathbf{T}_{\mathbf{DRj}}\cdot\frac{\mathbf{df}\left(\boldsymbol{\pi}_{\mathbf{inc}j}\right)}{\mathbf{d}\boldsymbol{\pi}_{\mathbf{inc}j}}\underline{\boldsymbol{\lambda}}_{2j}+\mathbf{k}_{\mathbf{E}j}^{-}\cdot\mathbf{T}_{\mathbf{DRj}}\cdot\frac{\mathbf{df}\left(\boldsymbol{\pi}_{\mathbf{inc}j}\right)}{\mathbf{d}\boldsymbol{\pi}_{\mathbf{inc}j}}\overline{\boldsymbol{\lambda}}_{5j}\end{aligned} \tag{58}$$

$$0\leq\lambda_{ij,t}\perp -f_i^{CS}\left(x_{j,t}^{CS}\right)\geq 0,\ i\in N_{ineq},\ j\in N_{CS} \tag{59}$$

where bold letters denote vector form (i.e. $\mathbf{x}=\left(x_1,...,x_t,...,x_T\right)$) and the operation $\mathbf{x}\cdot\mathbf{y}$ is to multiply vector $\mathbf{x}$ and $\mathbf{y}$ by multiplying corresponding elements. Matrix $\mathbf{A}$ is equal to

$$\mathbf{A}=\begin{bmatrix}1&0&\cdots&0\\-1&1&\ddots&\vdots\\\vdots&\ddots&\ddots&0\\0&\cdots&-1&1\end{bmatrix}_{T\times T}. \tag{60}$$

Equations (55)-(58) are the stationary constraints derived from the first order derivatives of Lagrange function with respect to the decision variables in the lower-level problem. Besides, (59) is the complementary slackness condition and can be linearized by the big-M method as follows:

$$\lambda_{ij,t}\geq 0,-f_i^{CS}\left(x_{j,t}^{CS}\right)\leq Mu_{ij,t},\lambda_{ij,t}\leq M\left(1-u_{ij,t}\right) \tag{61}$$

where M is a sufficiently large number, and $u_{ij,t}$ is the binary variable.

Incorporating these equilibrium constraints into the upper-level problem, the bilevel problem is transformed into a single-level mathematical program with equilibrium constraints (MPEC) problem [21]. Besides, by the strong duality theorem, the bilinear term of the cost from CSs in (10) can be replaced by the sum of a set of linear and nonlinear terms as follows:

$$\begin{aligned}&\sum_{j\in N_{CS}} \boldsymbol{\pi}_{\mathbf{j}}^{\mathbf{Ru}}\mathbf{Ru}_{\mathbf{j}}^{\mathbf{CS}} + \boldsymbol{\pi}_{\mathbf{j}}^{\mathbf{Rd}}\mathbf{Rd}_{\mathbf{j}}^{\mathbf{CS}} - \boldsymbol{\pi}_{\mathbf{j}}^{\mathbf{P}}\mathbf{P}_{\mathbf{j}}^{\mathbf{CS}} \\ &= \sum_{j\in N_{CS}} \left[\begin{array}{l} \nu_{2j}\left(E_j^+(T) - E_j^-(1)\right) + \left(\overline{\boldsymbol{\lambda}}_{2j} - \underline{\boldsymbol{\lambda}}_{2j}\right)^{\mathbf{T}} \mathbf{P}_{\mathbf{un},j} + \left(\overline{\boldsymbol{\lambda}}_{5j} - \underline{\boldsymbol{\lambda}}_{5j}\right)^{\mathbf{T}} \left(\mathbf{E}_j^+ - \mathbf{A}^{-1}\mathbf{C}_j\right) \\ + \underbrace{\left(\mathbf{k}_{\mathbf{Ej}}^- \cdot \mathbf{T}_{\mathbf{DRj}} \cdot \underline{\boldsymbol{\lambda}}_{\mathbf{5j}} + \mathbf{k}_{\mathbf{Pj}}^+ \cdot \mathbf{T}_{\mathbf{DRj}} \cdot \overline{\boldsymbol{\lambda}}_{\mathbf{2j}} + \mathbf{k}_{\mathbf{Pj}}^- \cdot \mathbf{T}_{\mathbf{DRj}} \cdot \underline{\boldsymbol{\lambda}}_{\mathbf{2j}}\right)^{\mathbf{T}} \mathbf{f}\left(\boldsymbol{\pi}_{\mathbf{incj}}\right)}_{\text{duality term of incentive program}} \end{array}\right]. \end{aligned} \tag{62}$$

where $\mathbf{C}_j = \left[E_{j,1}, 0, \ldots, 0\right]^T$.

C. Tractable reformulation

Observing the duality term of incentive program in Eq. (62), the tractability is strongly related to the fitting function $f(\boldsymbol{\pi}_{\mathbf{inc}})$. Thereby, in this Subsection, the tractable reformulation will be discussed in detail.

Lemma 1. If willingness fitting function $f(x)$ is differentiable, which also satisfies

$$f(x) = \exp\left(\int \frac{1}{kx^{\alpha}} dx\right) \tag{63}$$

where *k* is the positive coefficient and $\alpha \in \{0,1\}$. Then, Eq. (62) can be transformed into the convex function in its domain and the primal bilevel problem can be transformed into the tractable single-level problem.

Proof. Firstly, according to (58), $f(\boldsymbol{\pi}_{\mathbf{inc}})$ is differentiable. By strong duality theorem, multiplying (58) by $k(\boldsymbol{\pi}_{\mathbf{inc}})^{\alpha}$ with $\alpha = 1$, (58) can be rewritten as

$$\begin{aligned}&\pi_{inc}^{\max} \mathbf{1}^{\mathbf{T}} \overline{\boldsymbol{\lambda}}_{6j} + \left[\mathbf{T}_{\mathbf{DRj}} \cdot \left(\mathbf{f}\left(\boldsymbol{\pi}_{\mathbf{incj}}\right) + \frac{\mathbf{df}\left(\boldsymbol{\pi}_{\mathbf{incj}}\right)}{\mathbf{d}\boldsymbol{\pi}_{\mathbf{incj}}} \boldsymbol{\pi}_{\mathbf{incj}}\right)\right]^T \boldsymbol{\pi}_{\mathbf{incj}} \\ &= \left(\mathbf{k}_{\mathbf{P}j}^+ \cdot \mathbf{T}_{\mathbf{DRj}} \cdot \frac{\mathbf{df}\left(\boldsymbol{\pi}_{\mathbf{incj}}\right)}{\mathbf{d}\boldsymbol{\pi}_{\mathbf{incj}}} \overline{\boldsymbol{\lambda}}_{2j} + \mathbf{k}_{\mathbf{P}j}^- \cdot \mathbf{T}_{\mathbf{DRj}} \cdot \frac{\mathbf{df}\left(\boldsymbol{\pi}_{\mathbf{incj}}\right)}{\mathbf{d}\boldsymbol{\pi}_{\mathbf{incj}}} \underline{\boldsymbol{\lambda}}_{2j}\right)^T \boldsymbol{\pi}_{\mathbf{incj}} + \left(\mathbf{k}_{\mathbf{E}j}^- \cdot \mathbf{T}_{\mathbf{DRj}} \cdot \frac{\mathbf{df}\left(\boldsymbol{\pi}_{\mathbf{incj}}\right)}{\mathbf{d}\boldsymbol{\pi}_{\mathbf{incj}}} \underline{\boldsymbol{\lambda}}_{5j}\right)^T \boldsymbol{\pi}_{\mathbf{incj}} \end{aligned} \tag{64}$$

Comparing the right-hand side of (64) and the duality term of incentive program in (62), one can have

$$f(\pi_{inc}) = k\pi_{inc} \frac{df(\pi_{inc})}{d\pi_{inc}} \Rightarrow f(x) = Cx^{\frac{1}{k}}. \tag{65}$$

Thus, by substituting the left-hand side of (64) with the duality term of (62), (62) is transformed into a convex function. Similarly, when $\alpha = 0$, $f(\boldsymbol{\pi}_{\mathbf{inc}})$ becomes the exponential function and the conclusion is still valid. ■

Therefore, the polynomial or exponential functions can be chosen as the fitting function of $f(\boldsymbol{\pi}_{\mathbf{inc}})$ and we denote the first kind of willingness as **Type 1 willingness**. In other words, Lemma 1 provides a way of convex curve fitting for drivers' willingness. However, if the statistics of practical drivers' willingness is closer to a concave function, the fitting function of (63) might be inapplicable. To overcome this, we give the following Lemma 2.

Lemma 2. Let piecewise linear concave function $f(\boldsymbol{\pi}_{\mathbf{inc}})$ be a concave curve fitting of drivers' willingness, and define a new decision variable $f_{inc,t}$ as optimal willingness and satisfy

$$f_{inc,t} \le f\left(\pi_{inc,t}\right). \tag{66}$$

Then, the primal bilevel problem can be transformed into the tractable single-level problem.

Proof. Firstly, the willingness function $f_{inc,t} = f\left(\pi_{inc,t}\right)$ is modified by inequality (66) and the illustration of (66) is shown in Fig. 4. Since $f(\boldsymbol{\pi}_{\mathbf{inc}})$ is a piecewise linear concave function, (66) can be denoted by a group of linear constraints as:

$$a_k \pi_{inc,t} + b_k - f_{inc,t} \ge 0 : \lambda_{7k,t}, \forall t, k \tag{67}$$

$$\lambda_{ev}^{\max} - f_{inc,t} \ge 0, f_{inc,t} \ge 0 : \overline{\lambda}_{8,t}, \underline{\lambda}_{8,t}, \forall t \tag{68}$$

where $a_1 > ... > a_{N_{seg}} \geq 0$ and $b_{N_{seg}} > ... > b_1 \geq 0$. $k \in N_{seg}$, $N_{seg} := \{1,...,N\}$ and N is the number of segments. Then, the KKT conditions of (58) can be modified as

$$\mathbf{T_{DRj}} \cdot \mathbf{f_{incj}} + \overline{\boldsymbol{\lambda}}_{6j} - \underline{\boldsymbol{\lambda}}_{6j} - \sum_{k \in N_{seg}} a_{k,j} \boldsymbol{\lambda}_{7\mathbf{k},j} = 0 \tag{69}$$

$$\mathbf{T}_{\mathbf{DR}j} \cdot \boldsymbol{\pi}_{\mathbf{inc}j} + \overline{\boldsymbol{\lambda}}_{8j} - \underline{\boldsymbol{\lambda}}_{8j} + \sum_{k \in N_{seg}} \boldsymbol{\lambda}_{7\mathbf{k},j} = \mathbf{k}_{\mathbf{P}j}^{+} \cdot \mathbf{T}_{\mathbf{DR}j} \cdot \overline{\boldsymbol{\lambda}}_{2j} + \mathbf{k}_{\mathbf{P}j}^{-} \cdot \mathbf{T}_{\mathbf{DR}j} \cdot \underline{\boldsymbol{\lambda}}_{2j} + \mathbf{k}_{\mathbf{E}j}^{-} \cdot \mathbf{T}_{\mathbf{DR}j} \cdot \underline{\boldsymbol{\lambda}}_{5j}. \tag{70}$$

Further, by the strong duality theorem, the bilinear term of the cost from CSs in (10) can be rewritten as

$$\sum_{j \in N_{CS}} \boldsymbol{\pi}_{\mathbf{j}}^{\mathbf{Ru}} \mathbf{Ru}_{\mathbf{j}}^{\mathbf{CS}} + \boldsymbol{\pi}_{\mathbf{j}}^{\mathbf{Rd}} \mathbf{Rd}_{\mathbf{j}}^{\mathbf{CS}} - \boldsymbol{\pi}_{\mathbf{j}}^{\mathbf{P}} \mathbf{P}_{\mathbf{j}}^{\mathbf{CS}} = \sum_{j \in N_{CS}} \begin{bmatrix} \nu_{2j} \left(E_j^{+}(T) - E_j^{-}(1) \right) + \pi_{incj}^{max} \mathbf{1}^{\mathbf{T}} \overline{\boldsymbol{\lambda}}_{6j} \\ + \left(\overline{\boldsymbol{\lambda}}_{2j} - \underline{\boldsymbol{\lambda}}_{2j} \right)^{\mathbf{T}} \mathbf{P}_{\mathbf{un},j} + \left(\overline{\boldsymbol{\lambda}}_{5j} - \underline{\boldsymbol{\lambda}}_{5j} \right)^{\mathbf{T}} \left(\mathbf{E}_j^{+} - \mathbf{A}^{-1} \mathbf{C}_j \right) \\ + \mathbf{1}^{\mathbf{T}} \sum_{k \in N_{seg}} b_{kj} \boldsymbol{\lambda}_{7kj} + \lambda_{ev}^{max} \mathbf{1}^{\mathbf{T}} \overline{\boldsymbol{\lambda}}_{8j} + \left(2\mathbf{T}_{\mathbf{DR}j} \cdot \mathbf{f}_{\mathbf{inc}j} \right)^{\mathbf{T}} \boldsymbol{\pi}_{\mathbf{inc}j} \end{bmatrix}. \tag{71}$$

Consequently, the proposed bilevel problem can be transformed into the tractable single-level optimization. Besides, we denote the second kind of concave willingness ($f_{inc,t} \leq f\left(\pi_{inc,t}\right)$) as **Type 2 willingness**. ■

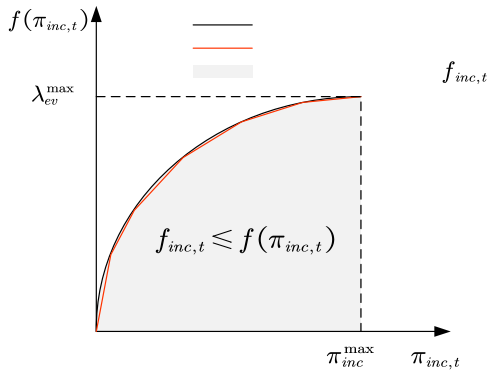


Fig. 4. Linearization and approximation of concave willingness function.

Thereby, based on **Lemma 1** and **2**, the tractable reformulation of the primal bilevel optimization can be achieved for both convex and concave responsivity functions of $f\left(\pi_{inc,t}\right)$.

VI. Case Studies

In this section, the proposed approach considering drivers' response willingness is evaluated by conducting case studies on the IEEE 33-bus and the 123-bus DNs. The proposed approach is implemented in Python environment and solved by Gurobi 9.5.1 on a PC with an Intel Core i5-8250U CPU@ 1.80GHz and 8GB of RAM.

A. IEEE 33-Bus system set up

The modified IEEE 33-bus system is with 2 DGs, 1 photovoltaic (PV) and 4 CSs, which is shown in Fig. 5. The maximum charging prices for CSs employ the time of use (TOU) price mechanism. TOU mechanism is a kind of price-based DR program and offers some flexibility to EV drivers for charging since it is set up based on the wholesale energy price[14]. The proposed incentive-based DR program is compatible with the TOU mechanism, which can be used to further explore the scheduling potentials of EVs.

The expected wholesale energy, TOU charging prices and up/down reserve prices are derived from the electricity market of China Southern Power Grid Co., Ltd., in Guangdong Province, which are shown in Fig. 6. The uncertainties of wholesale market are included in the ADN's model by Eq. (7), utilizing scenarios for wholesale energy and reserve prices. To reduce the computational burden, the optimal scenario reduction method in [37] is adopted in this paper. A set of scenarios are reduced to a scenario tree and the reduced scenarios represent the most probable scenarios with different probabilities. In this paper, 30 scenarios for the wholesale market uncertainties are reduced to three typical scenarios to obtain the expected dispatching results in the ADNs. The probabilities of $\gamma_{1,2,3}$ are 0.4, 0.37, and 0.23, respectively. The data of the market prices and the results of scenario reduction can be referred to [38].

The forecast nodal load and PV generation data are obtained from the State Grid Corporation of China. Besides, the uncertainties of PV and loads are described by $\xi_t^{L,net}$, with its standard deviation given by $10\%\mathbf{P}_t^{L,net}$. The risk level of DSO is $\varepsilon = 5\%$.

The real EV charging data contains the charging service records in four downtown CSs in Guangdong Province in 2022, which is obtained from Southern Power Grid Corporation [38]. The EV data include t_a, t_d, $p_{i,\max}$ and the charging energy demand. Based on EVA model of (29)-(33), the day-ahead EVA hourly operation boundary can be forecasted.

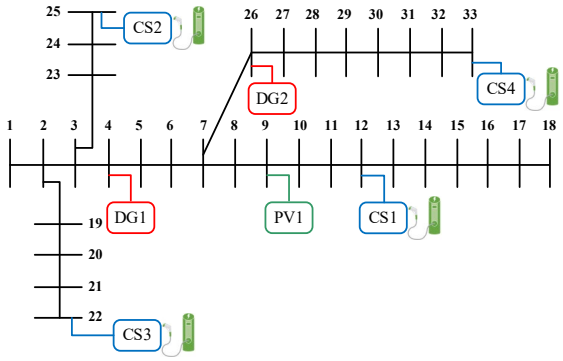


Fig. 5. The modified 33-bus distribution network.

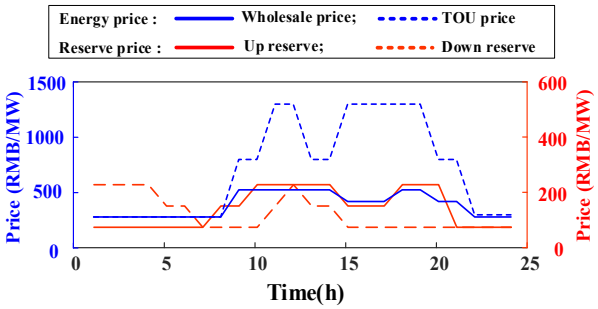


Fig. 6. Prices in the wholesale energy market and the reserve service market.

A.1 Performance of proposed method with different $f\left(\pi_{inc}\right)$

The proposed model falls into a scenario-based scheduling optimization and typical fitting functions of $f\left(\pi_{inc}\right)$ are chosen for CSs: i). Linear function for CS1. ii). Concave quadratic function for CS2 and CS3. The concave function is approximated by the piecewise linear concave function with three segments. iii). Convex quadratic function for CS4. Meanwhile, considering the charging fees of 50kWh EVs are generally around 30 RMB in China, the corresponding $\pi_{inc}^{\max}$ of drivers at different CSs are set equal to 3 RMB in this subsection. $\lambda_{ev}^{\max}$ is set equal to 90%. The impact of $\lambda_{ev}^{\max}$ and $\pi_{inc}^{\max}$ will be discussed in the next subsection.

The proposed optimization model considering drivers' responsivity uncertainties (**Model 1**) is compared with **Model 2:** the optimization model with only linear responsivity of drivers (without considering diversified responsivity); **Model 3:** the optimization model with no drivers' participation. The comparative results of different agents' cost are summarized in Table II. In addition, the day-ahead optimization results of the proposed model are shown in Fig. 7–Fig. 11. Specifically, several observations that shed light on the proposed model and the utilization of EV resources are listed as follows.

Firstly, comparing the proposed **Model 1** with **Model 3** in Table II, as for the cost of DSO, the energy cost decreases by 2931 RMB (3.6% reduction), the reserve cost decreases by 3642 RMB (19.1% reduction), and the total cost decreases by 2129 RMB (3.7% reduction). In the meanwhile, the total costs of CSs decrease by 1046–3272 RMB (10.4%-33.9% reduction), depending on different responsivity of drivers and locations of CSs. Besides, EV drivers can also gain profits of 853–1198 RMB (9.6% charging costs on average) from participating into DR. Thereby, the proposed program can achieve the win-win situation among the DSO, CSs and EV drivers.

In addition, compared with **Model 2** only considering linear responsivity of drivers, the proposed **Model 1** with diversified responsivity is more practical for CS operator to set incentives and make scheduling decisions. Specifically, the costs of CS1 in **Model 1** and **2** are similar to each other since the type of responsivity at CS1 is linear. However, the overestimation of drivers' participation at CS4 will violate the operation constraints of CS4 and further lead to security issues to ADNs.

Secondly, observing upper-level results in Fig. 7 and Fig. 8. It can be observed in Fig. 7(a),(b) that the proposed approach enhances the secure operation of the ADN under the uncertainties of RESs and loads. Besides, the reserve capacities from 9 a.m. to 9 p.m. are mainly provided by the wholesale market and CSs in Fig. 7(c). This is because DGs operate in their maximum ratings during high energy price period. In Fig. 8, the energy and reserve prices for different CSs vary due to different responsivity of EV drivers, CS locations, and available reserve capacities. In addition, the optimized charging price is lower than TOU and the reserve price is larger than zero, which indicates that system participants can benefit from the collaborative operation compared to the uncoordinated charging with only TOU mechanism.

TABLE II
COMPARISON RESULTS OF COST OF DIFFERENT AGENTS (UNIT: RMB)

	DSO	CS1	CS2	CS3	CS4
Model 1					
Total cost	56075	9996	9646	8129	10094
Energy cost	79867	11132	11380	9381	10937
Reserve cost	19038	-2334	-2659	-2105	-2008
Incentives for drivers	-	1198	924	853	1164
Model 2					
Total cost	56415	10091	10049	8525	8907
Energy cost	79803	11156	11399	9550	9960
Reserve cost	18677	-2205	-2497	-2065	-2098
Incentives for drivers	-	1140	1147	1040	1045
Model 3					
Total cost	58204	12556	12918	10658	11140
Energy cost	82798	12556	12918	10658	11140
Reserve cost	22680	0	0	0	0

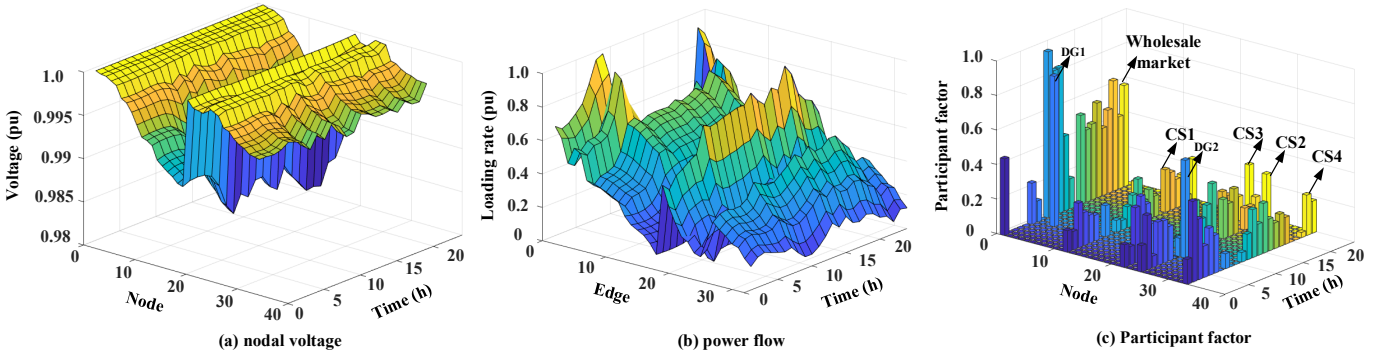


Fig. 7. The results of network operation.

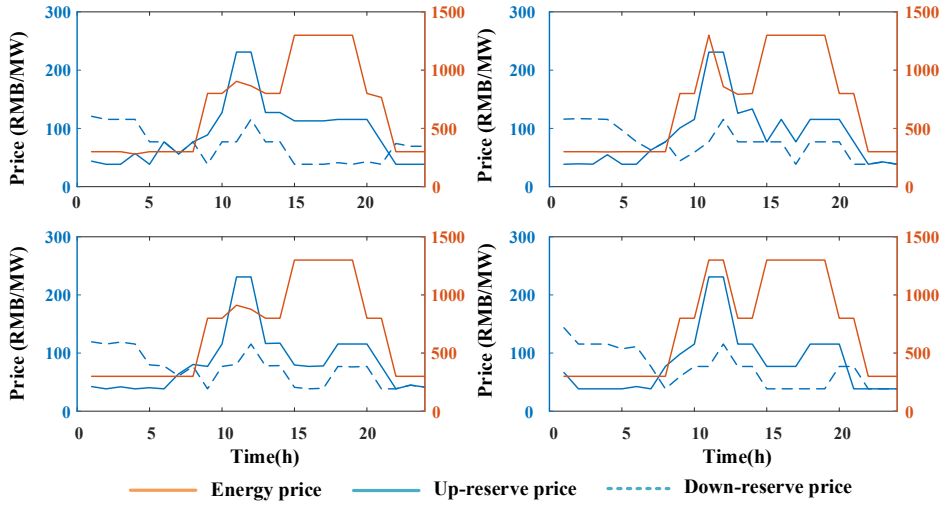


Fig. 8. The optimal energy and reserve prices for different CSs.

Thirdly, observing lower-level results in Fig. 9, the incentive prices for EV drivers are high when the charging and reserve prices are high. Therefore, CSs can provides more auxiliary services for the ADN and earn more profits. Besides, the willingness functions have strong influences on incentive prices. More specifically, as for **Type 1 willingness**, incentive prices are set higher since the increment of willingness is large when $\pi_{inc,t} > 2$. In contrast, the incentive prices of **Type 2 willingness** are mostly around 2 RMB since the increment of drivers' responsivity is small when $\pi_{inc,t} > 2$.

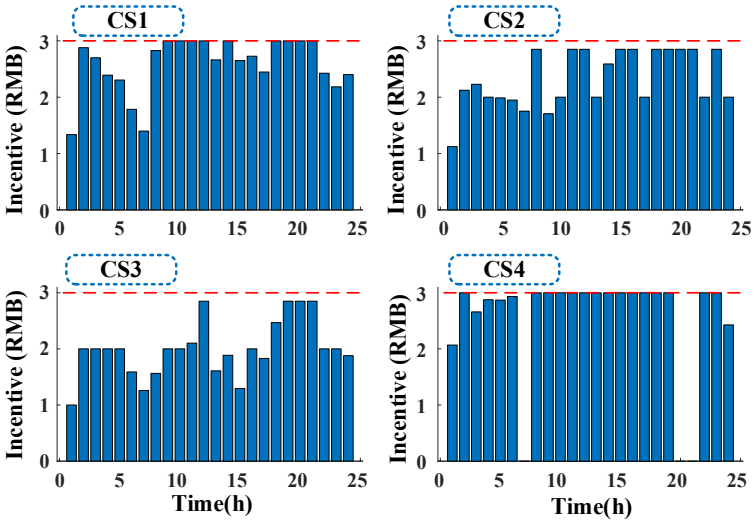


Fig. 9. The optimal incentive prices for EV drivers at four CSs.

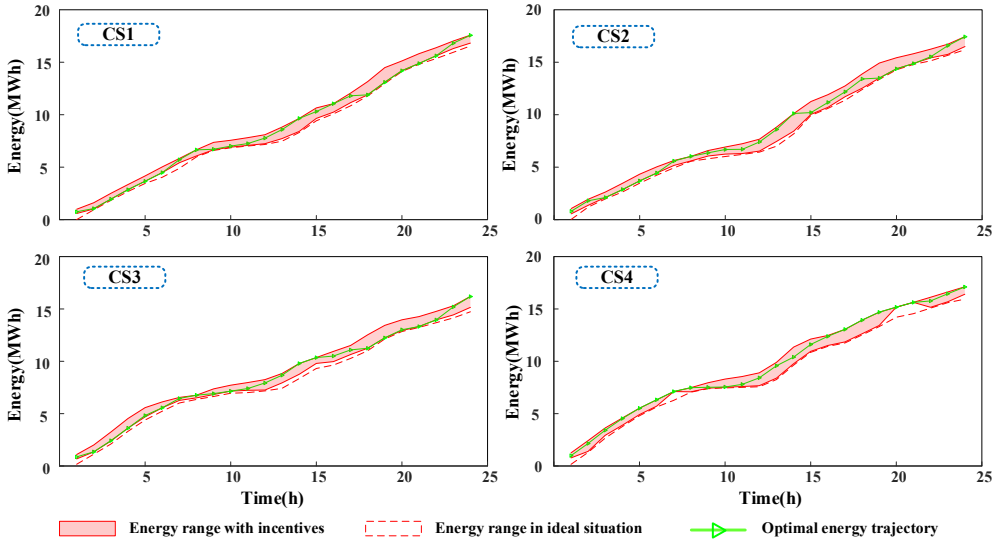


Fig. 10. The optimized energy trajectory within feasible ranges at four CSs.

Finally, the optimized energy and power trajectories in Fig. 10 and Fig. 11 are jointly determined by optimized energy/reserve prices, charging demands of EVs and willingness types of drivers. For instance:

i). The energy trajectories at four CSs increase to the upper bounds when the energy price is low during period of 1:00–8:00. The corresponding power trajectories have the similar features in Fig. 11 and CSs can provide some reserve capacities to earn higher profits.

ii). The energy trajectories in Fig. 10 become flat and even decrease to lower bounds when the energy price is high during 10:00–12:00 and 15:00–20:00, which is consistent with the power trajectories in Fig. 11.

iii). As for CS4 with the quadratic willingness, CS4 doesn't set incentives for EVs during 8:00-9:00 and 20:00-22:00. Thereby, CS4 charges EVs to the upper energy boundary in advance before peak charging price period and then it can reduce the charging cost. For other CSs, they have similar charging trajectories and collaboratively optimize the ADN scheduling (i.e., consume abundant PV generation at midday and avoid line congestion issues during the peak load period).

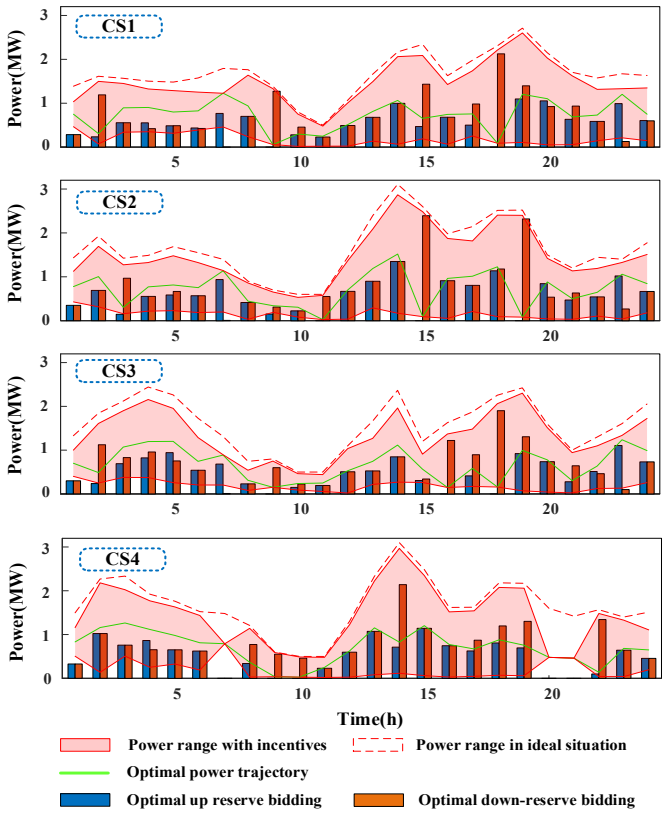


Fig. 11. The optimized power and reserve bidding plans at four CSs.

A.2 Influences of $\lambda_{ev}^{\max}$ *on costs of the DSO, CSs and Drivers*

This subsection further conducts sensitivity analysis on the varying $\lambda_{ev}^{\max}$ impact of the proposed model. $\pi_{inc}^{\max}$ is fixed at 3 RMB. The results are illustrated in Fig. 12.

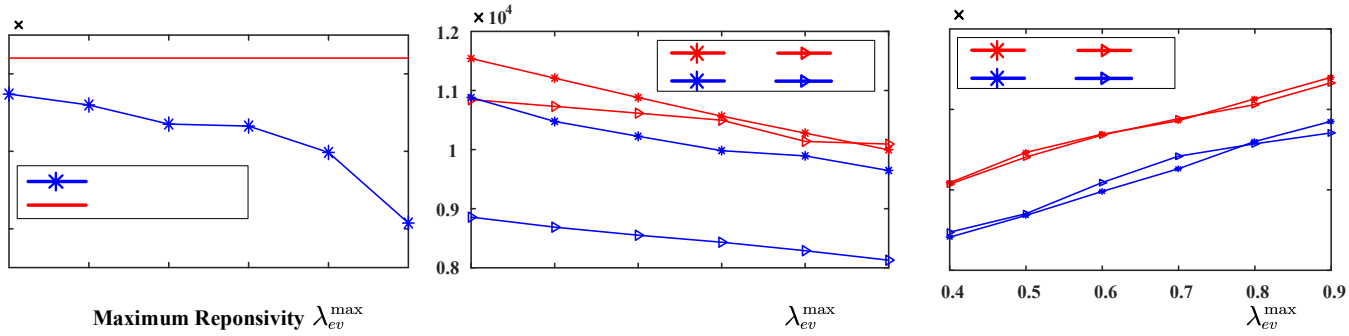


Fig. 12. Influences of $\lambda_{ev}^{\max}$ on costs of DSO, CSs and EVs.

Observing Fig. 12, the costs of DSO and CSs decrease gradually with the increase of $\lambda_{ev}^{\max}$ which indicates that the higher $\lambda_{ev}^{\max}$ of EVs can help both DSO and CSs reduce costs by increasing the areas of EVA operation range. Meanwhile, the incentives for drivers also increase with higher $\lambda_{ev}^{\max}$. Quantitively, with the increase of $\lambda_{ev}^{\max}$ from (40% to 90%), the costs of DSO, CSs and EV drivers reduce from 0.7% to 3.7%, 10.5% to 20% and 3.3% to 9.6%, respectively. In other words, the revenues of DSO, CSs and EV drivers all have positive correlation with $\lambda_{ev}^{\max}$ and the higher $\lambda_{ev}^{\max}$ of drivers can bring benefits to all system participants.

A.3 Influences of $\pi_{inc}^{\max}$ on costs of DS, CSs and Drivers

The practical $\pi_{inc}^{\max}$ of EV drivers are also important to CSs since the revenues of all stakeholders are sensitive to it. $\lambda_{ev}^{\max}$ is fixed at 90% and the influences of $\pi_{inc}^{\max}$ on the optimization results are shown in Fig. 13.

Firstly, the cost of DSO increases gradually with higher $\pi_{inc}^{\max}$. This is because the cost of auxiliary service provided by CSs becomes larger when $\pi_{inc}^{\max}$ increases.

Secondly, as for the costs of CSs in Fig. 13 (b), we divide them into **Type 1** and **2 willingness** to analyze. The costs of CS1 and CS4 with **Type 1 willingness** have approximately linear correlation with $\pi_{inc}^{\max}$. In contrast, the costs of CS2,3 with **Type 2 willingness** have approximately convex quadratic correlation with $\pi_{inc}^{\max}$. One reasonable illustration is that CSs collaboratively reach the global optimum of the system. Meanwhile, the optimal responses of CSs with the same type of willingness should be consistent with the change of $\pi_{inc}^{\max}$. Finally, EV drivers with **Type 2 willingness** have higher willingness to participate in the DR program and thereby, CS2 and 3 have lower costs when $\pi_{inc}^{\max}$ changes.

Finally, as seen in Fig. 13 (c), the concave quadratic function can be used to describe the correlation between the incentives for drivers and $\pi_{inc}^{\max}$. In other words, when $\pi_{inc}^{\max}$ is smaller than the critical point (the critical point is around 4 RMB in the studied case), CSs can afford more EV incentives to provide auxiliary services for the ADN and decrease their total costs. When $\pi_{inc}^{\max}$ is larger than the critical point, CSs can't gain profits from incentive increase and then decrease EV incentives to reduce their total costs. In a nutshell, if the practical $\pi_{inc}^{\max}$ of EV drivers is smaller than the critical point, EVs have considerable potentials of scheduling flexibility. Otherwise, the scheduling flexibility of EVs might be fractional.

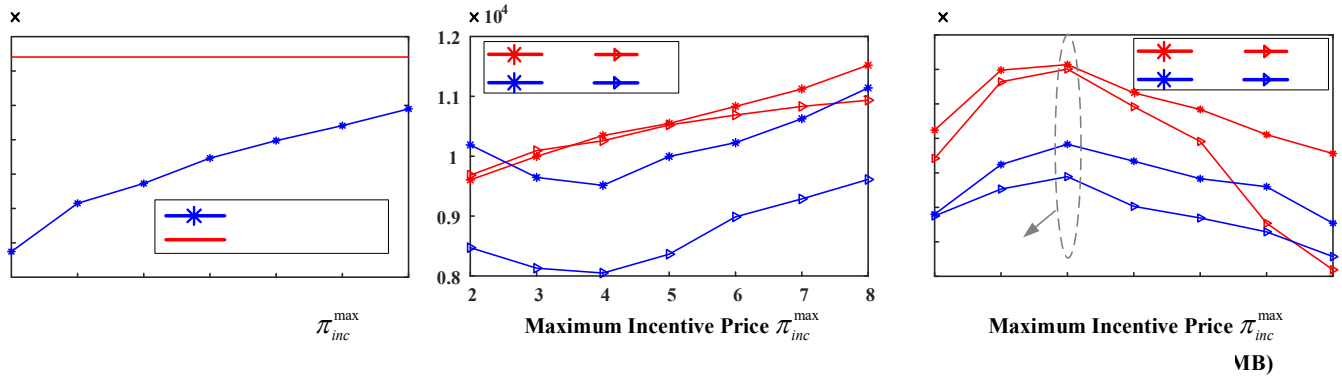


Fig. 13. Influences of $\pi_{inc}^{\max}$ on costs of the DSO, CSs and EVs.

B. Scalability test under the modified IEEE 123-bus system

The effectiveness and scalability of the proposed approach is further evaluated under the modified balanced IEEE 123-bus DN [38]. There are twelve DGs at bus-8, 21, 26, 42, 55, 57, 72, 81, 87, 91, 97 and 105, and two PVs at bus-43 and 59. Besides, two CSs at buses (17, 22) are with linear willingness curves, three CSs at buses (38, 50, 82) are with concave willingness curves, and one CS at bus (104) is with the convex quadratic willingness curve. The settings of drivers' responsivities, uncertainties of loads and wholesale market, and the risk levels are the same as those in the IEEE-33 DN case.

TABLE III
COST RESULTS OF DIFFERENT AGENTS IN THE IEEE 123-BUS SYSTEM (UNIT: RMB)

	DSO	CS1	CS2	CS3	CS4	CS5	CS6
The cost of different agents with the proposed method							
Total cost	159818	10533	10370	8980	9078	9661	10151
Energy cost	193130	11714	11850	10420	10700	11532	11078
Reserve cost	33986	-2302	-2616	-2293	-2534	-2847	-2080
Incentives for drivers	-	1120	1137	852	911	976	1152
The cost of different agents without considering EVs' participation							
Total cost	163212	12556	12918	10658	11140	12711	11058
Energy cost	197059	12556	12918	10658	11140	12599	11210

Reserve cost	37195	0	0	0	0	0	0

The comparative results are summarized in Table III. Firstly, the cost of DSO via the proposed strategy are reduced significantly compared to the uncoordinated charging method. Therefore, the proposed strategy can enlarge the scheduling flexibility of EVs and help DSO unlock the EVs dispatching potentials. Besides, CSs can also reduce their charging costs, provide auxiliary services for DSO and make optimal decisions of EV incentives for different willingness based on the proposed method. Finally, the computation time of the proposed approach in the IEEE 123-bus system is less than 600s, which is acceptable for the day-ahead scheduling in the large-scale power system.

VII. Conclusions and Future Works

In this paper, a day-ahead bilevel optimal scheduling strategy for DSO with multiple CSs considering drivers' willingness is proposed. Based on the consumer psychology model, different willingness functions of EV drivers are established and integrated in the DR model of CSs. Then, by using the KKT optimality conditions, strong duality theorem and linearization method, the bilevel optimization is transformed into a single-level tractable problem.

Furthermore, numerical studies have been conducted to substantiate the following results:

1) The proposed incentive program can decrease the costs of DSO, CSs, EV drivers by 3.7%, 20%, 9.6% in total on average, which achieves the win-win situation among system participants.

2) The characteristic parameters of drivers' responsivity are analyzed. With the increase of $\lambda_{ev}^{\max}$ from (40% to 90%), the costs of DSO, CSs and EV drivers reduce from 0.7% to 3.7%, 10.5% to 20% and 3.3% to 9.6%, respectively.

3) The proposed model is implemented in IEEE 123-bus DN to verify its scalability and computational efficiency, with the computation time less than 600s for the day-ahead ADN optimal scheduling.

In this work, the uncertainties of loads, responsivity of EV drivers and wholesale markets are modeled by CCP and scenario-based SO, respectively. The limitation of the proposed model is that the prior knowledge of distribution of these uncertainties is required which necessitates sufficient historical data. Thereby, the distributionally robust model can be explored in the future works which is more robust and only requires limited data. In addition, ongoing efforts are developing distributed frameworks for the coordinated operation between DSO and EVAs which can achieve privacy protection.

Acknowledgement

This work was supported in part by the National Natural Science Foundation of China under Grant 52277107, in part by the Shenzhen Science and Technology Program, Grant No. WDZC20220808143010001 and WDZC20231127152020001.

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